The Volumetric Buoyancy–Occupancy Interface Model: A Three-Phase Picture of the Dark Sector, with Falsifiable Threads and Honest Limits

TZPID Gold Spine Series II, Paper XVI of XX

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Abstract

Paper XII left the dark-energy term favoured but un-derived. This paper builds the mechanical model from which that term is meant to come, and marks the boundary between what the model earns and what it asserts. We formalize a three-phase occupancy field in which ordinary matter is the finite-viscosity interface condensate between a topological-current phase (candidate dark matter) and a breathing-volume phase (candidate dark energy), with constraint $\phi_b + \phi_T + \phi_E = 1$ and effective volumes $V_i = \int_{S^3} \phi_i \, dV$. Matter forms as the sech^2 domain wall between immiscible phases, and the enclosure seeks global volumetric equilibrium $\int_V [\mathbf{J} \times \mathbf{B} - \rho\, \boldsymbol{\Omega} \times \mathbf{v} + \rho\, \mathbf{g}_{\mathrm{eff}} + \nabla P_E + \mathbf{f}_{\mathrm{int}} - \nabla \cdot \boldsymbol{\tau}] \, dV = 0$, balanced at Elsasser criticality $\Lambda_{\mathrm{E}} = B^2/(\mu_0 \rho \eta \Omega) \approx 1$. The model's strength is that it trades metaphor for handles: the Elsasser number, magnetic-helicity conservation $H_M = \int_V \mathbf{A} \cdot \mathbf{B} \, dV$, the Hodge decomposition as a substance-versus-residual diagnostic, and the domain-wall soliton are each established physics. We isolate the two falsifiable threads—the $\Lambda_{\mathrm{E}} \approx 1$ attractor, testable now against dynamos, and the conjecture that the breathing pressure ∇P_E derives $w(a)$—and state the limits without softening: a symbol collision on Λ, an R^{-4} scaling that is radiation-like rather than dark-energy-like, an unmet set of dark-matter bars (rotation curves, the Bullet cluster, the CMB third peak), and a near-tautological global balance law. The model is retained on that footing, and the $\nabla P_E \to w(a)$ thread is discharged in Paper XVII.

1 Introduction: from a favoured term to a mechanism

The Spartan Dawn test favoured a dynamical dark-energy term $F(a)$ but supplied no mechanism; it fitted the equation of state rather than predicting it. A mechanism is owed. This paper assembles one from physical handles rather than metaphor, and spends as much effort marking the model's limits as stating its content. The result is not a finished theory of the dark sector but a structured proposal with two falsifiable exposures and four explicit weaknesses.

2 The three-phase construction

Let ϕ_b, ϕ_T, ϕ_E be the occupancy fractions of ordinary matter, the topological-current phase, and the breathing-volume phase, with

$$\phi_b + \phi_T + \phi_E = 1, \qquad V_i(t) = \int_{S^3} \phi_i \, dV. \tag{1}$$

The phases are immiscible (a quartic exclusion energy $\mathscr{U} = \sum_{i<j} \chi_{ij}\phi_i\phi_j$), and ordinary matter forms as the interfacial domain wall $\phi_b(s) \propto \mathrm{sech}^2(s/\ell)$, the standard kink, with $s = 0$ the interface. Globally the enclosure seeks volumetric equilibrium,

$$\int_V [\mathbf{J}\times\mathbf{B} - \rho\,\boldsymbol{\Omega}\times\mathbf{v} + \rho\,\mathbf{g}_{\mathrm{eff}} + \nabla P_E + \mathbf{f}_{\mathrm{int}} - \nabla\cdot\boldsymbol{\tau}]\,dV = 0, \tag{2}$$

"locally twisting, globally balanced." Ordinary matter is the visible film where the topological-current drive and the breathing-volume pressure cannot co-occupy a state channel.

3 The two falsifiable handles

Proposition 3.1 (Elsasser attractor)**.** *The dimensionless Elsasser number $\Lambda_{\mathrm{E}} = B^2/(\mu_0\rho\eta\Omega)$ has an empirical attractor $\Lambda_{\mathrm{E}} \approx 1$ for active dynamos, observed in the band $0.1 < \Lambda_{\mathrm{E}} < 10$, giving the equilibrium field $B_\star = \sqrt{\mu_0\rho\eta\Omega}$. This is testable now: rotating conductive systems—laboratory dynamos, planetary cores, stars—should self-organize toward $\Lambda_{\mathrm{E}} \approx 1$.*

Conjecture 3.2 (Breathing pressure yields the equation of state)**.** *If the breathing pressure gradient ∇P_E in Eq.* (2) *can be derived from the nested-enclosure dynamics, it should yield $w(a)$. The target is $w_0 \approx -0.6$, $w_a \approx -1.4$ (Paper XII). Deriving these, rather than fitting them, converts the model from accommodation to prediction.*

The Hodge decomposition $\omega = d\alpha + \delta\beta + H_k$ with inversion $A = \delta(\Delta^{-1}B)$ is a genuine diagnostic: it separates an observed field into gradient, curl, and topological-residue parts, posing the operational question "is the dark component a missing substance or a topological residual?" Magnetic helicity $H_M = \int_V \mathbf{A}\cdot\mathbf{B}\,dV$ supplies a conserved invariant, conjecturally combined as $H_M + \lambda_g^{-1} H_{\mathrm{ac}} = \mathrm{const}$, the recycling law.

4 Honest limits

Remark 4.1 (Symbol collision). The Elsasser number $\Lambda_{\mathrm{E}} \approx 1$ and the cosmological constant $\Lambda \approx 10^{-52}\,\mathrm{m}^{-2}$ are unrelated quantities sharing a glyph. A dynamo-balance result says nothing about dark energy; the shared symbol must never be read as an identification.

Remark 4.2 (R^{-4} is radiation-like). The nested-mode density $\rho_n \sim R_n^{-4}$ is the dilution law of radiation, the opposite of dark energy's near-constant density ($w \approx -1$, Paper XII). The R^{-4} law is correct for the vacuum-mode density $\rho_\Lambda = \hbar c/R^4$ only because there R is one fixed, enormous radius—not a shrinking shell. Conflating the two is the model's most seductive error.

Remark 4.3 (Dark-matter bars unmet). A topological-current dark matter must reproduce flat rotation curves, the collisionless Bullet-cluster separation, the CMB third-peak baryon-to-dark ratio, and structure growth. Until ϕ_T clears those bars, it is a name, not a model.

Remark 4.4 (Near-tautology). Volumetric equilibrium $\int_{S^3} \mathbf{F}\,dV = 0$ on a boundaryless manifold is essentially momentum conservation—true but almost free. Predictive content must come from the terms of Eq. (2), not from the fact that they sum to zero.

5 Verification status and demarcation

The components are independently established: the Elsasser number and its ≈ 1 attractor (magnetohydrodynamics); magnetic-helicity conservation (ideal MHD); the Hodge decomposition (differential topology); the sech^2 kink (ϕ^4 theory). What is not verified is the assembly into a model of the dark sector—the hypothesis, not a checked result.

Established physics: the Elsasser number and the dynamo band; magnetic-helicity conservation; the Hodge decomposition; the domain-wall profile.

Framework synthesis: the three-phase occupancy assignment; the volumetric-equilibrium master equation (2); and Elsasser criticality as the interface's neutral-buoyancy attractor.

Predictions: (i) rotating conductive systems converge to $\Lambda_{\mathrm{E}} \approx 1$; (ii) matter-forming regions sit at finite-viscosity neutral-buoyancy shells; (iii) high-field systems should show a reduced apparent dark-matter requirement through magnetic/topological stress.

6 Limits of the construction

Beyond the four remarks, the model's largest debt is predictive: it organizes the dark sector rather than computing any dark-sector observable. The Elsasser thread concerns dynamos, not cosmology directly; the dark-matter assignment is unbacked by the four bars; and the breathing-pressure thread is, at this paper's close, still a conjecture. The model is retained as a scaffold with two exposures, not an established account. Its honesty is its present virtue: every component is established, every assembly claim is flagged, and the one assembly claim that can be made to compute—the dark-energy equation of state—is handed to Paper XVII.

7 Conclusion: a scaffold with two exposures

The Volumetric Buoyancy–Occupancy model is the most mechanistically complete version of the dark-sector picture in the series, because it trades metaphor for handles—Elsasser balance, helicity conservation, Hodge diagnostics, domain-wall condensation. Its value is concentrated in two falsifiable threads and bounded by four explicit limits. The thread that matters most is Conjecture 3.2: that the breathing pressure ∇P_E can derive the dark-energy equation of state the Spartan Dawn fit could only accommodate. Paper XVII takes that thread to its end.

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